

Dorset Software Airport Simulator: **User Guide**

This guide provides a comprehensive walkthrough of the **Dorset Software Airport Simulation** tool. Designed for digital consultancies and airport managers, this application models aircraft throughput and operational disruptions using high-efficiency **Discrete Event Simulation (DES)**.

Running the Simulation

There are 3 ways to run the application:

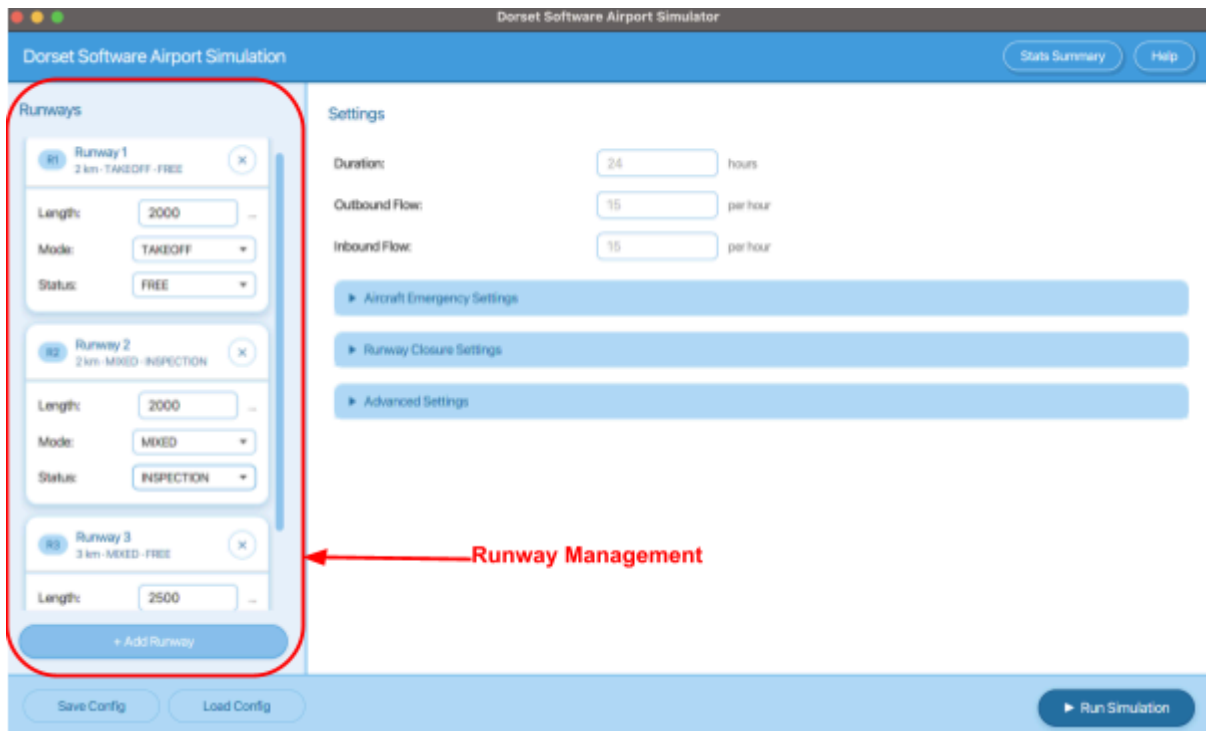
- **Command Line** - All users can run this application by downloading the application, then navigating to the folder
`\CS261-Software-Engineering-Aircraft-Sim-` in the command line.
Users then run the line `mvn javafx:run` to start the application.
- **Windows** - The user can run the file `AirportSimulation.exe` by double-clicking on the file icon on File Explorer. (Located in
`CS261-Software-Engineering-Aircraft-Sim-\AirportSimulation\AirportSimulation.exe`)
- **Mac** - The user can run the Application file `AirportSimulationMac`. (Located in
`CS261-Software-Engineering-Aircraft-Sim-\AirportSimulationMac`).

1. The Start Screen: Configuring Your Scenario

The **Configuration Stage** allows you to define the physical and operational parameters of your airport.

Runway Management

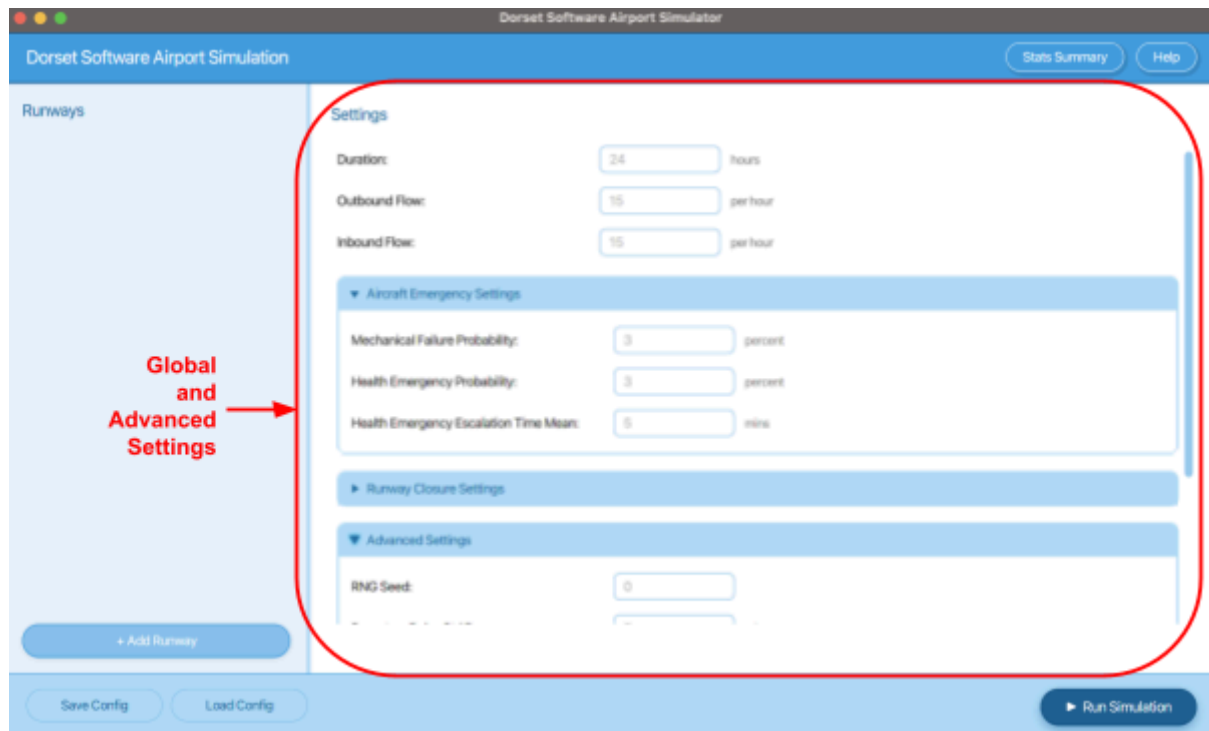
- **Adding Runways:** Click the **" + Add Runway "** button to define your airport layout, with support for up to **10 runways**.
- **Runway Parameters:** Each runway is individually configurable:
 - **Length:** Define the physical length (between 500m and 4000m), which determines the time between an aircraft entering and exiting a runway.
 - **Mode:** Select **Landing**, **Take-off**, or **Mixed Mode** (handling both arrivals and departures).
 - **Status:** Set the initial operational state, such as **Available**, **Runway Inspection**, **Snow Clearance**, or **Equipment Failure**.



Global & Advanced Settings

Organize your simulation by adjusting traffic flow and underlying statistical probabilities:

- **Simulation Duration:** Specify the length of simulation time the model will run for before it terminates, measured in hours.
- **Traffic Flow:** Set the **Inbound** and **Outbound Flow** rates, which determine the average number of aircraft arriving or departing per hour.
- **Aircraft Emergency Settings:** Configure the likelihood of disruptions:
 - **Mechanical Failure Probability:** Percentage chance of technical issues occurring.
 - **Health Emergency Probability:** Likelihood of a passenger health crisis occurring.
 - **Health Emergency Escalation Time Mean:** The average time before a mild health issue becomes a severe emergency.
- **Runway Closure Settings:** Adjust the mean duration and probability for **Inspections**, **Snow Clearance**, and **Equipment Failures**.
- **Advanced Settings:**
 - **RNG Seed:** Enter a specific number to ensure the simulation is **deterministic**, allowing you to reproduce the exact same random events across multiple runs.
 - **Low Fuel Threshold:** Define the point (default 10 minutes) at which an aircraft must divert or land immediately.
 - **Max Delay:** The maximum time an aircraft waits in the take-off queue before the flight is **cancelled**.



Configuration Tools

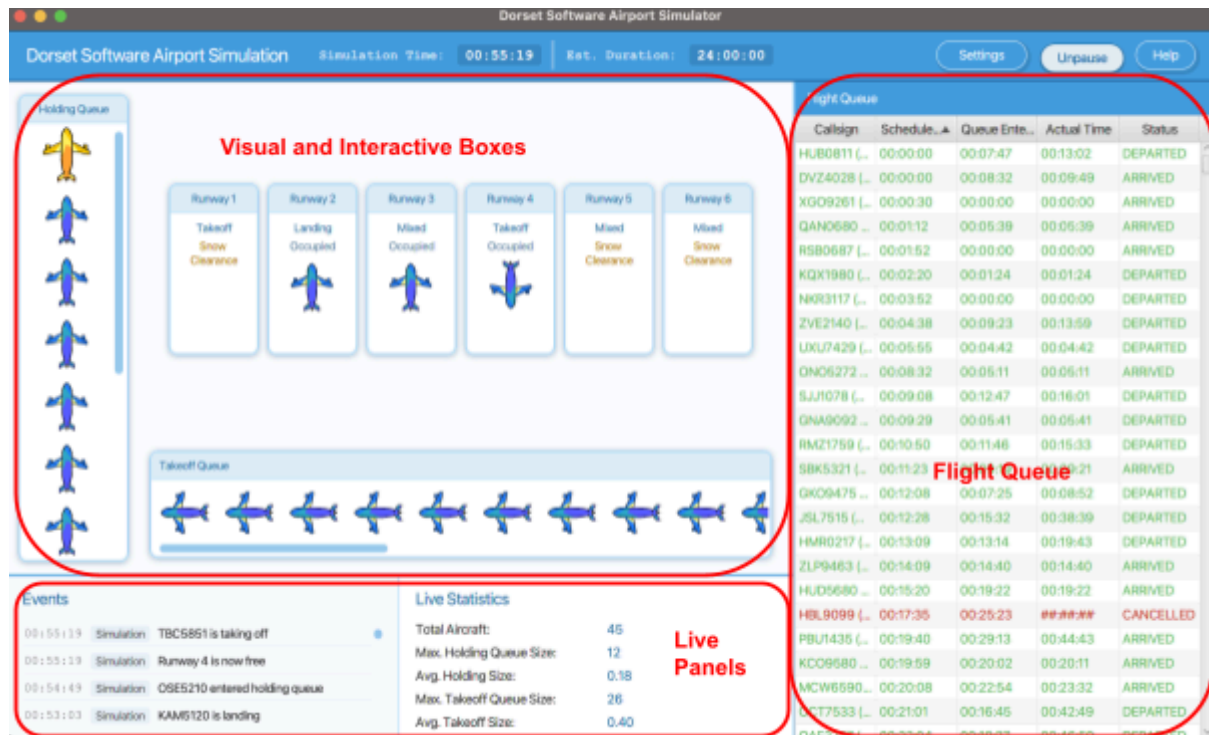
- **Save/Load Config:** Use the **"Save Config"** and **"Load Config"** buttons to retain your input parameters for future use or to share specific scenarios with other team members.
- **Help Button:** Click the **"Help"** button at the top right for a guided explanation of the configuration fields and navigation options.

2. Live Simulation: Real-Time Monitoring

Click **"Run Simulation"** to enter the **Execution Stage**, where the airport state is visualized in real-time.

The Flight Queue & Live Panels

- **Flight Queue Panel (Right):** Displays a live timetable of all aircraft, including **Callsign, Scheduled Time, Queue Entry Time, Actual Time** and current **Status** (Arrived/Departed).
- **Live Statistics (Bottom Center):** Displays real-time updates on **Average Queue Size, Max Queue Size, and Total Aircrafts**.
- **Event List (Bottom Left):** A chronological feed of significant simulation events, such as runway closures or emergency declarations.



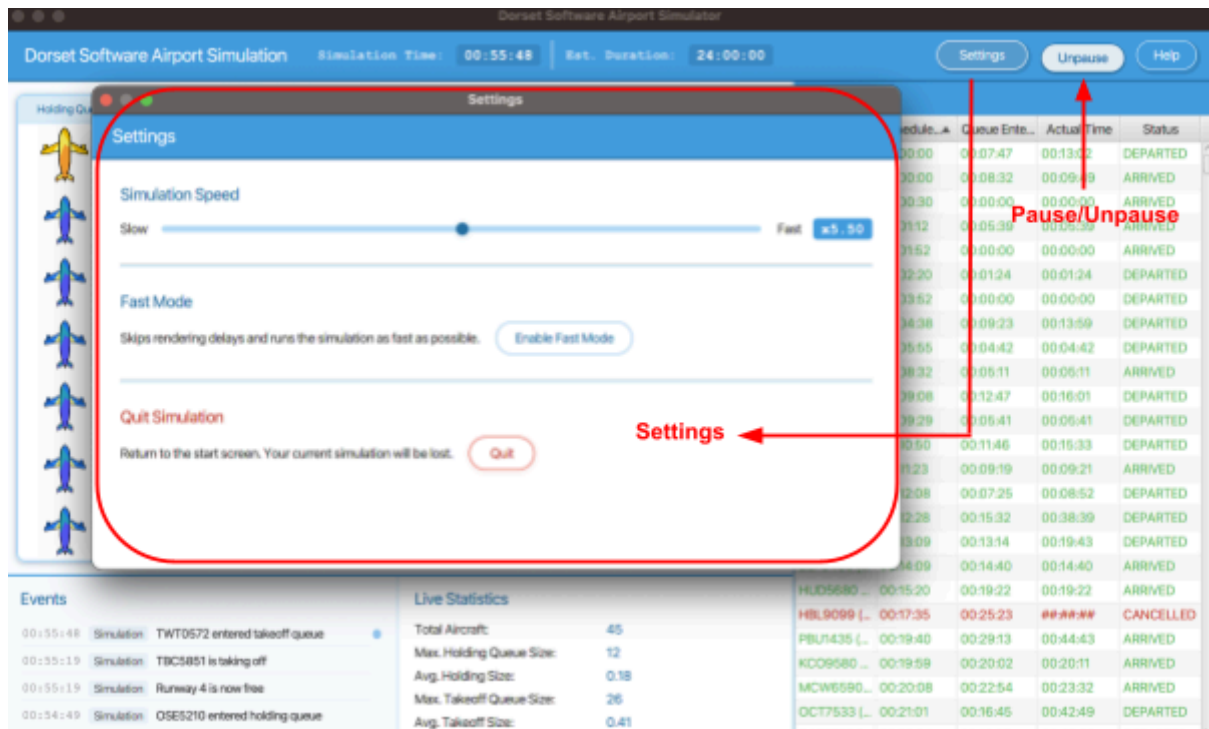
Visual Cues & Interaction

- **Queue Boxes:**
 - **Holding Queue:** Displays arriving aircraft arranged by priority (emergencies first, then arrival time).
 - **Take-off Queue:** Displays aircraft waiting for a departure slot.
 - **Expansion:** Clicking on a queue box reveals detailed information for every aircraft currently inside, such as priority and fuel levels.
- **Runway Boxes:** Each box shows the runway's current mode and occupancy.
 - **Color Coding:** Aircraft icons change color based on emergency status: **Red** (Severe), **Yellow** (Mild), and **Blue** (None).
 - **Orientation:** Aircraft are oriented **Upwards** for landing and **Downwards** for take-off.

Simulation Controls

- **Pause & Edit:** Click **Pause** to freeze the simulation. While paused, click on any runway to change its **Mode** or **Status** mid-run to react to congestion.
- **Speed Adjustment:** Use the speed slider in settings to slow down or speed up the simulation playback.
- **Fast Mode:** Computes the simulation instantly, jumping straight to the results. Can help generate the result of a **10,000 hour simulation** in less than **5 seconds**.
- **Help Button:** Provides contextual information regarding the live view, queue interactions, and runway states.

- **Quit Simulation:** Exit the current run to return to the start screen or close the application.

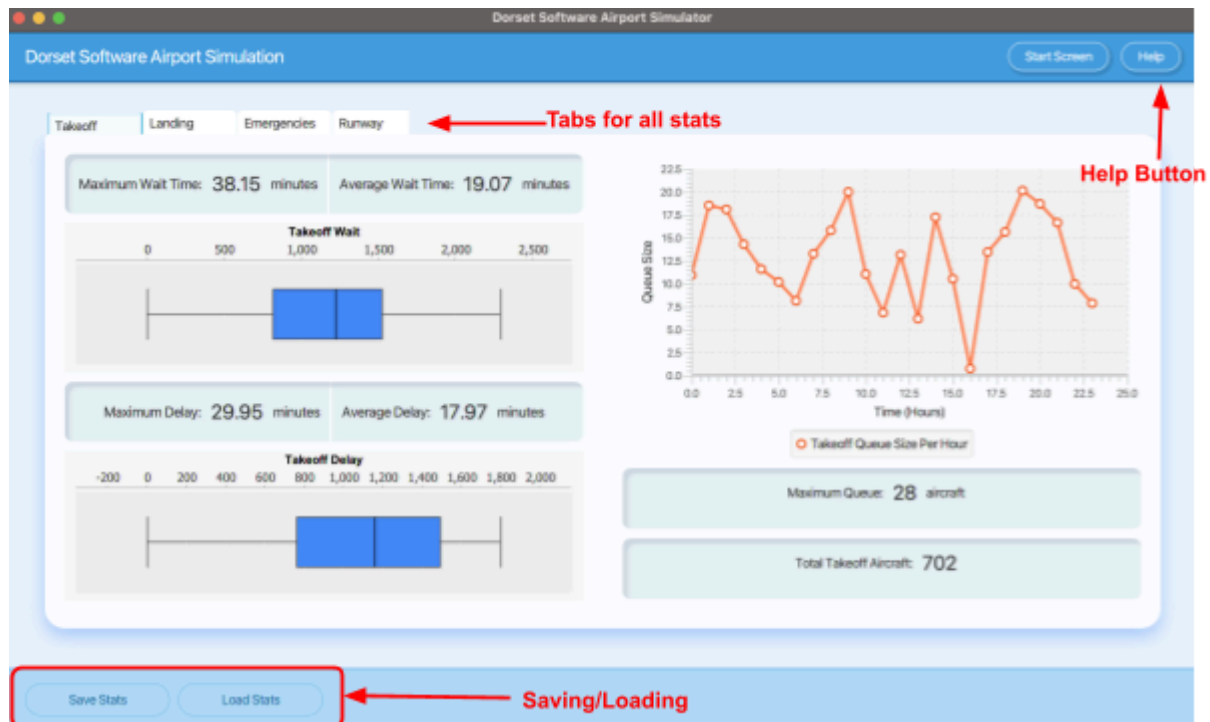


3. Statistics & Analysis: Post-Simulation Report

After the simulation finishes, the **Analysis Stage** provides aggregated performance data. Use the **tabbed interface** at the top to scroll through different categories of data.

Tab 1: Take-Off Stats

- **Max Queue Size & Averages:** Displays the peak number of planes in the queue and the average time spent waiting to take off.
- **Box Plots:** Visualizes the distribution of wait times and delays (Min, 25th percentile, Median, 75th percentile, and Max). This helps managers understand if wait times are consistently spaced or if outliers are causing significant issues.
- **Queue Size Line Graph:** Models how congestion changes hourly throughout the day, showing the impact of mid-simulation runway closures.



Tab 2: Landing Stats

- **Landing Analysis:** Follows the same format as take-off, providing **Max/Average hold times, delays**, and **Box Plots** focusing on the efficiency of the landing process and the size of the holding pattern.

Tab 3: Aircraft (Emergencies & Risks)

- **Cancellations & Diversions:** Total counts and hourly rates for aircraft that could not take off within the max delay or land before running low on fuel.
- **Emergency Pie Chart:** Breaks down the specific types of aircraft emergencies encountered, such as mechanical failure or passenger health.
- **System Stability Score:** A unique reliability metric (0–100) representing how many aircraft successfully landed or departed. A higher score indicates a safe and dependable configuration.

Tab 4: Runway Stats

- **Closure Summary:** Lists the total number of closures caused by **Inspections**, **Equipment Failures**, or **Snow**.
- **Runway Mode Pie Chart:** Compares the number of runways set to different modes. A toggle allows you to see the difference between the **Initial** and **Final** setup after mid-run edits.
- **Closure Ratio:** Summarizes total time lost to closures, calculated by the duration between a closure and the subsequent open event.

Analysis Tools

- **Save/Load Stats:** Use the buttons at the bottom of the screen to save your results or load previous runs to compare performance between different airport configurations.
- **Start Screen Button:** Click the **"Start Screen"** button at the top right to go directly back to the configuration page and try a different simulation setup.
- **Help Button:** Offers an explanation of the various charts, box plots, and stability metrics displayed on the page.

4. Troubleshooting & Input Validation

Issue	Visual Indicator / Feedback	Solution
Input Out of Range	A red error message or prompt appears.	Ensure values are within accepted bounds.
Missing Input Values	An alert prompt asks if you want to continue.	Enter a valid number or click "Continue" to use the system's pre-configured default values.
Runway Limit Exceeded	An error message is displayed when clicking "+ Add Runway".	The simulation is capped at 10 runways.

Future Development

For any maintenance and future development of the code, JavaDocs have been provided.

This can be accessed via the folder

`\CS261-Software-Engineering-Aircraft-Sim-\docs` in the application files.